

Name _____ Period _____

Skill 27.1 Exercise 1

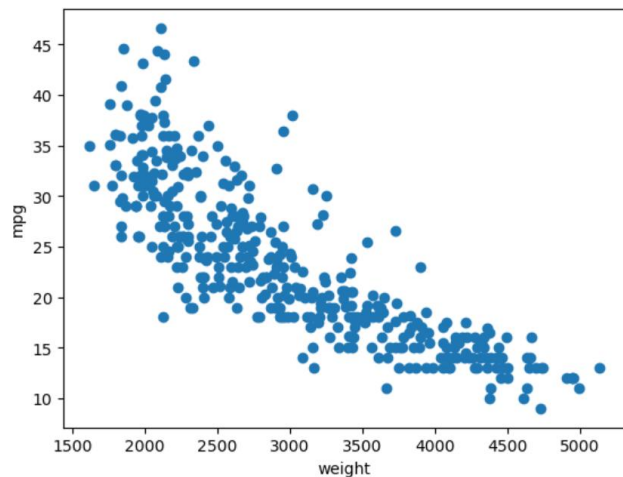
Refer to the *cars* dataframe below,

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	car name
0	18.0	8	307.0	130	3504	12.0	70	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	ford torino

Complete the spaces below to make a scatter plot to see if there is an association between *mpg* and *weight*.

```
plt.scatter(x = cars.weight, y = cars.mpg)
plt.xlabel('cylinders')
plt.ylabel('mpg')
plt.show()
```

Below is the actual plot



Based on the scatter plot, does vehicle weight appear to be associated with fuel efficiency (mpg)? Describe the direction and strength of the association.

Yes, vehicle weight appears to be strongly associated with fuel efficiency (mpg). The scatter plot shows a clear negative relationship: as vehicle weight increases, mpg tends to decrease. The points follow a fairly consistent downward trend, indicating that heavier vehicles generally have lower fuel efficiency. Because the points are closely clustered around this downward pattern, the association appears strong and approximately linear.

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Skill 27.2 Exercise 1

Refer to the cars DataFrame below.

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	car name
0	18.0	8	307.0	130	3504	12.0	70	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	ford torino

Use the `cov()` method to calculate the covariance matrix for *mpg* and *weight*.

```
cov_mat_weight_mpg = cars[['weight', 'mpg']].cov()
```

The actual matrix is as follows,

```

                weight      mpg
weight  717140.990526 -5505.211745
mpg      -5505.211745   61.089611

```

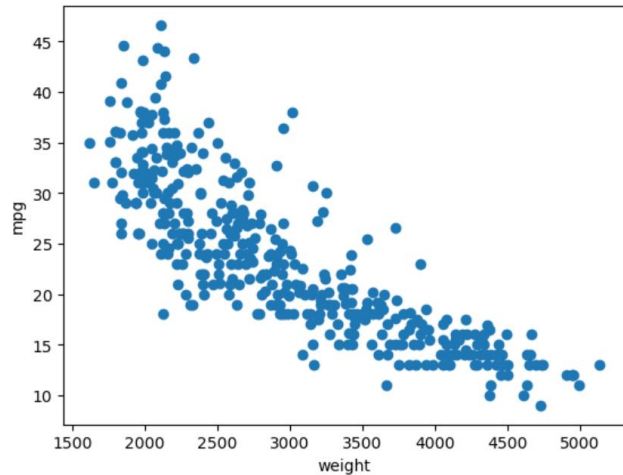
Identify which value in the matrix represents the covariance between weight and mpg. What does its sign tell you about their relationship?

The value is -5505.211745. The negative value tells us that there is a negative association between the variables. That is as the weight increases, the mpg goes down. Because we do not know the units, the magnitude is ambiguous.

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Skill 27.3 Exercise 1

Below is the plot of the mpg and weight.



Use the *pearsonr* function from *scipy.stats* to calculate the correlation between *mpg* and *weight* and print the result.

```
from scipy.stats import pearsonr

mpg_weight_corr = pearsonr(cars.weight, cars.mpg)
print(mpg_weight_corr[0])
```

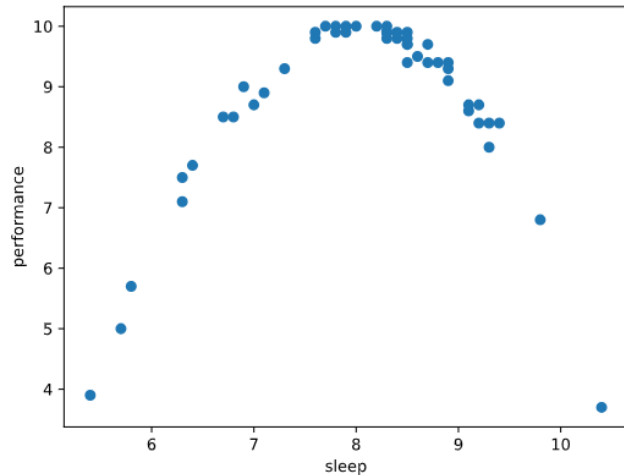
The actual value is -0.831740933244335. Is this value consistent with the scatter plot?

-0.83 indicates a strong negative linear relationship. This is consistent with the shape of the scatter plot

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Skill 27.4 Exercise 1

The plot below shows the relationship between hours of sleep and performance score on a task.



Describe the relationship between sleep and performance shown in the scatter plot.

The scatter plot shows an inverted U-shaped pattern, where performance increases as sleep increases up to about 8 hours, and then decreases as sleep continues to increase

Would correlation accurately reflect the strength of this relationship?

No, correlation would not accurately reflect the strength of this relationship. Correlation measures the strength of a linear relationship,

If the correlation were near zero, would that mean sleep and performance are unrelated? Why or why not?

No, a correlation near zero would not mean that sleep and performance are unrelated. It would only mean there is little to no linear relationship.